

**Artificial Intelligence Adoption in Higher Education Institutions in
Latin America:
Institutional Performance, Governance, Administrative Efficiency,
and Student Outcomes**

Discussion Paper

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Abstract

The rapid diffusion of artificial intelligence (AI) technologies across sectors of the global economy has prompted renewed scholarly attention to their transformative potential in higher education. In the Latin American context, where universities face compounding pressures of fiscal constraint, growing enrollment demand, and persistent socioeconomic inequalities, AI adoption presents both unprecedented opportunities and significant governance challenges. This discussion paper examines the multidimensional impact of AI adoption in Latin American higher education institutions (HEIs), with particular emphasis on institutional performance, administrative governance, operational efficiency, and student academic outcomes. Drawing on a systematic review of the emergent literature, we develop a conceptual framework that integrates institutional theory, technology acceptance models, and human capital perspectives. We further propose an empirical strategy based on a quasi-experimental panel data design that exploits variation in the timing and intensity of AI adoption across institutions and national systems. Our analysis suggests that while AI technologies hold substantial promise for improving administrative efficiency and personalizing learning experiences, their benefits are unevenly distributed across institutions, disproportionately favoring well-resourced universities in more developed national systems. Policy implications underscore the need for coordinated regional frameworks, targeted capacity-building investment, and attention to algorithmic equity in the design of AI-mediated educational tools.

Keywords: artificial intelligence, higher education, Latin America, institutional performance, administrative governance, student outcomes, educational technology, equity

1. Introduction

The last decade has witnessed an exponential proliferation of artificial intelligence technologies across virtually every domain of human activity. From predictive analytics in healthcare to natural language processing in legal services, AI systems have demonstrated the capacity to reorganize production processes, reallocate human capital, and fundamentally alter the nature of institutional decision-making (Acemoglu & Restrepo, 2022). Higher education, as both a producer and consumer of knowledge, occupies a particularly central position in this transformation. Universities are simultaneously sites where AI research is generated, actors who must govern the adoption of these tools within their own administrative and pedagogical structures, and institutions whose primary output—educated human capital—is being redefined by the skill demands of an AI-integrated economy (Selwyn, 2022).

In the Latin American context, these dynamics take on a distinctive character. The region's higher education landscape is marked by considerable heterogeneity: elite research universities coexist with under-resourced public institutions serving first-generation students; centralized national systems contrast with decentralized, market-oriented models; and digital infrastructure gaps between urban and rural areas create layered inequalities of access (Bernasconi, 2008; García de Fanelli, 2014). Against this backdrop, the question of how AI adoption affects institutional outcomes is not merely technical but profoundly political, touching on issues of equity, sovereignty, and the social function of the university.

The existing literature, while growing rapidly, remains fragmented across disciplinary traditions. Computer scientists have tended to focus on the development and technical evaluation of AI tools for education—so-called EdTech systems—without attending sufficiently to their institutional context (Holmes et al., 2021). Sociologists and education researchers have raised critical questions about algorithmic bias and the commercialization of educational data (Williamson, 2020), but have often lacked the quantitative tools to rigorously estimate causal effects. Economists of education have begun to bring quasi-experimental methodologies to bear on the question of technology adoption in schools and universities (Cristia et al., 2017; Beuermann et al., 2015), yet these studies have focused primarily on hardware access—computers and internet connectivity—rather than on the more recent wave of AI-enabled applications.

This paper aims to contribute to the emerging literature at the intersection of these perspectives. We pursue three objectives. First, we provide a structured review of what is currently known about AI adoption in Latin American HEIs, synthesizing evidence from both the region and comparable global contexts. Second, we develop a conceptual framework capable of organizing the multiple channels through which AI may affect institutional performance, governance, and student outcomes. Third, we propose an empirical research strategy suited to the data environment of Latin

American higher education systems, with attention to identification challenges and threats to validity.

The remainder of this paper is organized as follows. Section 2 presents a review of the relevant literature. Section 3 develops our conceptual framework. Section 4 describes the proposed empirical strategy. Section 5 discusses policy implications. Section 6 concludes.

2. Literature Review

2.1 AI in Global Higher Education: Emerging Evidence

The scholarly literature on AI in higher education has expanded dramatically since approximately 2018, coinciding with the mainstreaming of machine learning applications and, more recently, large language models. Reviews by Zawacki-Richter et al. (2019) and Chen et al. (2020) document the range of AI applications studied in academic contexts, encompassing intelligent tutoring systems, automated essay scoring, predictive analytics for dropout prevention, chatbot-mediated student support, and administrative automation. The preponderance of this work originates from institutions in the United States, the United Kingdom, China, and Australia, reflecting both the geographic concentration of EdTech investment and the availability of large-scale institutional data.

Evidence on the effectiveness of AI-based instructional tools is mixed. Studies of intelligent tutoring systems, such as those built on the Cognitive Tutor platform, have reported meaningful improvements in mathematics achievement in controlled settings (VanLehn, 2011). However, systematic reviews caution that effect sizes are highly variable and that publication bias likely inflates published estimates (Kulik & Fletcher, 2016). More recent research on AI-generated feedback in writing instruction has shown promising results in improving revision behavior, though concerns about surface-level feedback and the displacement of instructor engagement remain (Stevenson & Phakiti, 2019).

The use of predictive analytics for early warning and student success interventions has attracted considerable policy interest. Systems that flag students at risk of withdrawal or failure based on learning management system (LMS) engagement data and demographic characteristics have been implemented at scale in several U.S. institutions. Evaluations, however, suggest that without accompanying human advising interventions, predictive flags alone have limited impact on retention (Arnold & Pistilli, 2012). Concerns about racial and socioeconomic bias in prediction models add an important equity dimension to these findings (Kizilcec & Lee, 2022).

2.2 The Latin American Context

Latin American higher education systems have undergone substantial transformation since the 1990s, characterized by massification, privatization, and increasing differentiation (Levy, 2018). Regional enrollment rates have risen from approximately 18 percent in 1991 to over 50 percent in 2020, driven largely by expansion of private and distance-learning institutions (UNESCO-IESALC, 2021). This expansion has occurred against a backdrop of chronic underfunding of public universities, particularly in countries such as Brazil, Mexico, Argentina, and Colombia, where per-student public expenditure has stagnated in real terms (Ferreira et al., 2017).

Digital infrastructure in the region has improved substantially, yet significant gaps persist. Connectivity rates among higher education students exceed 80 percent in urban areas of countries such as Chile and Uruguay but fall below 50 percent in rural areas of Peru, Bolivia, and Paraguay (ITU, 2023). The COVID-19 pandemic, while devastating in its human costs, served as a forcing function for rapid digital adoption across the region, with universities compelled to shift teaching and administration online within weeks (Álvarez et al., 2021). This period accelerated institutional familiarity with digital platforms and created latent demand for more sophisticated AI-enabled tools.

Research specifically focused on AI adoption in Latin American HEIs remains sparse but is growing. Sánchez-Prieto et al. (2020) document adoption patterns of AI-enabled LMS features among faculty at Mexican and Spanish universities, finding that perceived usefulness and institutional support are the primary predictors of uptake. Rodríguez-Moreno et al. (2022) examine the use of chatbots for administrative query management in Colombian universities, documenting significant reductions in staff time spent on routine inquiries but noting student satisfaction challenges related to the impersonality of AI-mediated interactions. Cobo et al. (2023) provide a systematic bibliometric analysis of Latin American publications on educational AI, identifying Brazil, Mexico, and Colombia as the dominant producers of regional scholarship but noting that most published work is descriptive or normative rather than causal.

2.3 Institutional Performance and Governance

The governance of higher education institutions is increasingly recognized as a critical determinant of institutional performance outcomes, including research productivity, teaching quality, and financial sustainability (Dobbins et al., 2011). AI technologies introduce new governance challenges by changing information flows, creating new categories of institutional risk, and altering the distribution of decision-making authority between administrators, faculty, and external actors such as technology vendors (Williamson & Eynon, 2020).

The adoption of AI-driven management information systems—often branded as enterprise resource planning (ERP) systems with embedded AI modules—has been associated with improvements in budget management, enrollment forecasting, and human resource allocation in institutional case studies from the region (Herrera & López, 2021). However, the successful implementation of these systems requires substantial organizational change management and data governance capacity that many Latin American institutions lack (Rama, 2019). The literature on technology adoption in public organizations more broadly suggests that institutional inertia, legacy IT systems, and resistance from professional staff represent major barriers to AI integration (Wirtz et al., 2019).

2.4 Student Outcomes and Equity

The relationship between educational technology adoption and student outcomes has been extensively studied in the context of primary and secondary education in Latin America, with the

landmark evaluations of the One Laptop Per Child initiative in Peru and Uruguay providing important methodological templates (Beuermann et al., 2015; Cristia et al., 2017). These studies found modest or null effects of hardware access on test scores, while identifying positive effects on certain 21st-century skills and home educational activities.

At the higher education level, the causal evidence base is thinner, reflecting the greater complexity of the institutional environment and the challenges of obtaining administrative data from heterogeneous national systems. The available evidence suggests that the effects of technology adoption on student outcomes in Latin American HEIs are highly conditioned by prior academic preparation, socioeconomic background, and the quality of complementary inputs—particularly instructor training and institutional support services (Ferreira et al., 2017; Álvarez et al., 2021). This conditionality is theoretically consistent with a complementarity framework in which AI tools amplify rather than substitute for human educational inputs.

3. Conceptual Framework

3.1 Analytical Dimensions

We organize our conceptual framework around four interrelated analytical dimensions: (i) institutional performance, encompassing research output, graduation rates, and accreditation status; (ii) governance and organizational capacity, including decision-making structures, data management, and regulatory compliance; (iii) administrative efficiency, covering cost per student, processing times for administrative services, and staff allocation; and (iv) student outcomes, spanning academic achievement, retention, employability, and equity of access.

These dimensions are treated as simultaneously determined, reflecting the systemic nature of higher education institutions. AI adoption may affect administrative efficiency directly through the automation of routine tasks, while indirectly affecting student outcomes by reallocating saved administrative resources toward student-facing services. Similarly, governance capacity shapes the quality of AI implementation, which in turn conditions the effects on all other dimensions.

3.2 Theoretical Grounding

Our framework draws on three theoretical traditions. First, institutional theory (DiMaggio & Powell, 1983; Scott, 2008) provides a lens for understanding why and how universities adopt AI technologies, emphasizing the role of normative, coercive, and mimetic pressures. Universities in Latin America face coercive pressures from government digital modernization mandates, normative pressures from international accreditation bodies that increasingly expect evidence of technology integration, and mimetic pressures as peer institutions adopt AI tools in competitive prestige markets.

Second, technology acceptance models—particularly the Unified Theory of Acceptance and Use of Technology (UTAUT) developed by Venkatesh et al. (2003)—offer microfoundational accounts of how individual faculty, administrators, and students form attitudes toward and intentions to use AI tools. Factors including performance expectancy, effort expectancy, social influence, and facilitating conditions are posited to mediate the relationship between AI availability and actual use. In the Latin American context, the role of facilitating conditions—particularly institutional IT support and training—is likely to be especially important given the heterogeneity of digital readiness across institutions.

Third, human capital theory (Becker, 1964; Mincer, 1974) frames student outcomes in terms of the productivity of educational investments. From this perspective, AI tools represent a potential augmentation of the educational production function, capable of increasing the efficiency with which human capital is accumulated. However, consistent with the literature on complementarity in technology adoption (Acemoglu & Autor, 2011), we hypothesize that the returns to AI-enhanced

education will be concentrated among students who enter with higher levels of academic preparation and digital literacy.

3.3 Hypotheses

Drawing on the framework outlined above, we formulate four principal hypotheses to be evaluated in subsequent empirical work:

H1 (Administrative Efficiency): Institutions that adopt AI-enabled administrative systems will exhibit statistically significant reductions in cost per enrolled student and processing times for key administrative transactions, relative to non-adopting peers.

H2 (Governance Quality): Early and comprehensive AI adoption will be associated with improvements in institutional governance indicators, including budget execution rates and accreditation outcomes, conditional on baseline institutional capacity.

H3 (Student Outcomes—Average Effects): AI adoption in instructional contexts will be associated with modest positive average effects on student retention and graduation rates, with effect sizes concentrated in the first two years of enrollment.

H4 (Equity—Heterogeneous Effects): The benefits of AI adoption for student outcomes will be unevenly distributed, with larger positive effects for students from higher socioeconomic backgrounds and those enrolled in better-resourced institutions, consistent with a complementarity mechanism.

4. Proposed Empirical Strategy

4.1 Research Design

We propose a quasi-experimental research design exploiting variation in the timing and intensity of AI adoption across HEIs within and across Latin American national systems. The identification strategy relies on two sources of variation: (i) staggered adoption timing, in which institutions in different countries adopted specific AI tools at different points in time due to variation in national digital modernization policies; and (ii) intensity variation, in which institutions within the same country adopted AI tools at different rates depending on institutional size, funding levels, and prior technology investment.

The primary empirical specification takes the form of a difference-in-differences (DiD) estimator in a panel data setting. Let Y_{it} denote an outcome of interest (e.g., graduation rate, cost per student) for institution i in academic year t . Let AI_{it} be a continuous measure of AI adoption intensity, scaled between 0 and 1 based on a composite index described below. The baseline specification is:

$$Y_{it} = \alpha + \beta \cdot AI_{it} + \gamma \cdot X_{it} + \delta_i + \lambda_t + \varepsilon_{it}$$

where X_{it} is a vector of time-varying institutional covariates (enrollment, faculty-student ratio, public/private status, prior accreditation, research expenditure); δ_i are institution fixed effects absorbing time-invariant unobserved heterogeneity; and λ_t are year fixed effects capturing common shocks affecting all institutions simultaneously, including macroeconomic cycles and national education policy changes. Standard errors are clustered at the institution level to account for serial correlation in the error term.

To address the possibility of differential pre-trends—a threat to the parallel trends assumption underlying DiD estimation—we implement an event study specification that allows the effect of AI adoption to vary dynamically around the adoption date. Placebo tests using leads (pre-treatment periods) provide diagnostic evidence on the validity of the identification assumption.

4.2 AI Adoption Index

A central methodological challenge is the measurement of AI adoption. We propose constructing a composite AI Adoption Index (AAI) based on data gathered through an annual survey instrument administered to senior technology officers (Chief Information Officers or equivalent) at a stratified sample of HEIs across the region. The AAI captures four sub-dimensions: (i) administrative AI, including AI-enabled ERP modules, chatbots for student services, and predictive budgeting tools; (ii) instructional AI, encompassing AI-enabled LMS features, automated assessment, and intelligent tutoring systems; (iii) research AI, covering the use of AI tools in data collection,

literature synthesis, and manuscript preparation; and (iv) AI governance, reflecting the existence of institutional AI policies, ethics committees, and data governance protocols.

Each sub-dimension is scored on a Likert-type scale and aggregated using confirmatory factor analysis to produce a single continuous index, with appropriate attention to construct validity and reliability. The use of a multi-dimensional index, rather than a binary adoption indicator, allows for a richer characterization of the adoption landscape and enables analysis of heterogeneous effects across dimensions.

4.3 Data Sources

The proposed empirical strategy requires panel data at the institutional level spanning at least ten academic years (2015–2025), a period that encompasses both the pre-AI baseline and the rapid adoption phase accelerated by the COVID-19 pandemic. We identify four primary data sources: (i) national higher education information systems, including Brazil's Censo da Educação Superior, Mexico's ANUIES database, Colombia's SNIES, and Argentina's SPU data; (ii) the Latin American Quality Assurance Network (RIACES) database on accreditation outcomes; (iii) the AI Adoption Survey to be designed as part of this research program; and (iv) graduate tracer studies and labor market data from national statistical agencies.

A significant limitation is the incompleteness and inconsistency of administrative data across national systems, particularly for smaller and newer private institutions. We propose supplementing administrative records with institutional surveys and engaging national higher education ministries as partners in the data collection effort, following the model established by the Inter-American Development Bank's regional education data initiatives (Ferreira et al., 2017).

4.4 Threats to Validity and Robustness

Several threats to internal validity merit explicit discussion. First, selection into AI adoption may be non-random: institutions with greater prior capacity—larger endowments, stronger IT infrastructure, more technologically oriented leadership—may be both more likely to adopt AI early and more likely to exhibit favorable performance outcomes independently. Institution fixed effects address time-invariant selection, but time-varying confounding remains a concern. We will address this through rich covariate adjustment and by testing robustness to alternative specifications.

Second, the simultaneous adoption of multiple educational technologies during our study period—including LMS expansion, video conferencing infrastructure, and open educational resources—creates challenges for attributing observed effects specifically to AI. We address this by constructing separate indexes for AI and non-AI technology adoption and including both in our specifications.

Third, the relatively short panel and modest sample sizes in several national contexts may limit statistical power, particularly for estimating heterogeneous effects. We will conduct ex ante power calculations and pre-register our analysis plan to guard against selective reporting.

5. Policy Implications

5.1 Regional Coordination and Standards

The heterogeneous adoption landscape we have documented across Latin American HEIs underscores the need for regional coordination in AI governance. Currently, AI adoption in higher education is driven largely by individual institutional decisions and by the commercial strategies of global EdTech vendors, without adequate regional counterweights to ensure that adoption serves public educational purposes. We recommend that regional bodies including UNESCO-IESALC, the Inter-American Development Bank, and the Ibero-American States Organization for Education, Science and Culture (OEI) develop a Regional Framework for AI in Higher Education that articulates shared principles, interoperability standards, and mechanisms for cross-national data sharing.

Such a framework should address three critical areas: (i) data governance, including standards for the collection, storage, and use of student data by AI systems, with particular attention to the cross-border data flows that arise when universities license tools from multinational vendors; (ii) algorithmic transparency, requiring that AI tools used in consequential decision-making—such as student risk scoring or admissions support—be subject to independent audits; and (iii) equity standards, establishing minimum performance benchmarks for AI tools across socioeconomic and demographic groups to prevent the entrenchment of existing inequalities through algorithmic bias.

5.2 Capacity Building and Institutional Support

The evidence reviewed in this paper consistently indicates that the benefits of AI adoption are conditional on complementary institutional capacity, including technical infrastructure, faculty digital competence, and organizational change management skills. For many Latin American HEIs, particularly public universities in smaller countries and in rural or peripheral regions, these complementary conditions are absent. Policy interventions targeted exclusively at encouraging AI adoption without addressing capacity gaps will likely exacerbate rather than reduce institutional inequalities.

We recommend a tiered capacity-building program, funded through a combination of national education budgets and multilateral lending, that provides: (i) infrastructure grants for the modernization of institutional IT systems and connectivity; (ii) faculty development programs focused on AI literacy and pedagogical integration; (iii) technical assistance for the development of institutional data governance frameworks; and (iv) peer-learning networks that facilitate knowledge transfer between more and less advanced institutions within and across national systems.

5.3 Governing the Private EdTech Market

The Latin American EdTech market has attracted substantial private investment, with venture capital flows into the sector reaching record levels in the 2020–2023 period (HolonIQ, 2023). While this investment has expanded the availability of AI tools for HEIs, it has also introduced market dynamics that may not align with public educational objectives. Platforms that offer AI tools at subsidized rates in exchange for student data rights, or that lock institutions into proprietary ecosystems that impede interoperability, represent particular regulatory concerns.

National higher education regulators should develop specific provisions governing institutional contracts with AI tool providers, requiring transparency in data use agreements, ensuring portability of student data, and mandating disclosure of any commercial use of student behavioral or performance data. Procurement frameworks should be designed to favor open-source or locally developed AI solutions where functionally equivalent, building regional technological capacity while reducing dependence on foreign platforms.

5.4 Equity-Centered Implementation

Given our theoretical prediction—and the weight of global evidence—that AI adoption in education will tend to produce heterogeneous effects that favor more advantaged students, equity considerations must be central to implementation policy rather than treated as a secondary concern. Institutions should be required to conduct equity impact assessments prior to deploying AI tools in high-stakes contexts, and to monitor outcomes disaggregated by socioeconomic status, gender, ethnicity, geographic origin, and disability status.

Particular attention should be devoted to the risk of automation-induced disemployment among administrative staff in lower-wage positions, many of whom are themselves graduates of the institutions they serve. Transition support, reskilling programs, and consultation with labor organizations should be embedded in AI implementation processes to ensure that efficiency gains do not come at the cost of institutional employment and community well-being.

6. Conclusion

This discussion paper has examined the impact of artificial intelligence adoption on higher education institutions in Latin America through a structured review of the emerging literature, the development of a conceptual framework, and the specification of a rigorous empirical research strategy. Our analysis leads to several overarching conclusions.

First, AI technologies present genuine and significant opportunities for improving the efficiency, quality, and accessibility of higher education in Latin America. Administrative automation can reduce costs and improve service delivery; intelligent instructional tools can personalize learning and support at-risk students; and AI-enabled research infrastructure can accelerate knowledge production at institutions that have historically been disadvantaged in the global research system.

Second, the realization of these benefits is far from automatic. It requires sustained institutional investment in complementary capacities—infrastructure, human capital, and organizational governance—that are unevenly distributed across the region's highly heterogeneous higher education landscape. Without deliberate policy intervention, the adoption of AI is as likely to widen as to narrow existing institutional inequalities.

Third, the governance of AI in higher education represents a novel and undertheorized challenge. Existing frameworks for quality assurance, accreditation, and data protection were not designed with AI-mediated educational processes in mind. Updating these frameworks, at both national and regional levels, is an urgent policy priority that requires engagement not only from ministries of education and higher education institutions, but also from civil society, student organizations, and the research community.

Fourth, the empirical evidence base on AI in Latin American HEIs remains insufficient to support strong causal conclusions. The research agenda proposed in this paper—centered on a quasi-experimental panel data design exploiting variation in adoption timing and intensity—offers a credible path toward filling this gap. We invite collaborators across national systems to join in the development of the shared data infrastructure and institutional survey instrument that this agenda requires.

The transformation of higher education by artificial intelligence is underway, whether institutions are ready for it or not. The central task for researchers and policymakers is not to decide whether this transformation will occur, but to shape its direction, pace, and distribution in ways that serve the region's broader development objectives and the democratic mission of the public university.

References

- Acemoglu, D., & Autor, D. (2011). Skills, tasks and technologies: Implications for employment and earnings. In O. Ashenfelter & D. Card (Eds.), *Handbook of Labor Economics* (Vol. 4, pp. 1043–1171). Elsevier.
- Acemoglu, D., & Restrepo, P. (2022). Tasks, automation, and the rise in US wage inequality. *Econometrica*, 90(5), 1973–2016.
- Álvarez, H., Arias-Ortiz, E., Bergamaschi, A., López, A., Noli, A., Ortiz, M., Pérez, M., Rieble-Aubourg, S., Rivera, M. C., Scannone, R., Vásquez, M., & Viteri, A. (2021). La educación en tiempos del coronavirus: Los sistemas educativos de América Latina y el Caribe ante COVID-19. Inter-American Development Bank.
- Arnold, K. E., & Pistilli, M. D. (2012). Course Signals at Purdue: Using learning analytics to increase student success. In *Proceedings of the 2nd International Conference on Learning Analytics and Knowledge* (pp. 267–270). ACM.
- Becker, G. S. (1964). *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*. University of Chicago Press.
- Bernasconi, A. (2008). Is there a Latin American model of the university? *Comparative Education Review*, 52(1), 27–52.
- Beuermann, D. W., Cristia, J., Cueto, S., Malamud, O., & Cruz-Aguayo, Y. (2015). One laptop per child at home: Short-term impacts from a randomized experiment in Peru. *American Economic Journal: Applied Economics*, 7(2), 53–80.
- Chen, X., Xie, H., Zou, D., & Hwang, G. J. (2020). Application and theory gaps during the rise of artificial intelligence in education. *Computers and Education: Artificial Intelligence*, 1, 100002.
- Cobo, C., Rivera-Vargas, P., & Álvarez-Otero, S. (2023). Artificial intelligence in Latin American education: A bibliometric and thematic analysis. *Education and Information Technologies*, 28(4), 4123–4155.
- Cristia, J., Ibarrarán, P., Cueto, S., Santiago, A., & Severín, E. (2017). Technology and child development: Evidence from the One Laptop per Child Program. *American Economic Journal: Applied Economics*, 9(3), 295–320.
- DiMaggio, P. J., & Powell, W. W. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, 48(2), 147–160.

- Dobbins, M., Knill, C., & Vögtle, E. M. (2011). An analytical framework for the cross-country comparison of higher education governance. *Higher Education*, 62(5), 665–683.
- Ferreira, M. M., Avitabile, C., Botero Álvarez, J., Haimovich Paz, F., & Urzúa, S. (2017). *At a Crossroads: Higher Education in Latin America and the Caribbean*. World Bank.
- García de Fanelli, A. (2014). Rendimiento académico y abandono universitario: Modelos, resultados y alcances de la producción académica en la Argentina. *Revista Argentina de Educación Superior*, 6(8), 9–38.
- Herrera, C., & López, F. (2021). Implementación de sistemas ERP con inteligencia artificial en universidades colombianas: Factores críticos de éxito. *Revista de Educación Superior*, 50(197), 45–68.
- Holmes, W., Bialik, M., & Fadel, C. (2021). *Artificial Intelligence in Education: Promises and Implications for Teaching and Learning*. Center for Curriculum Redesign.
- HolonIQ. (2023). *Latin America EdTech 2023: Market Intelligence Report*. HolonIQ Analytics.
- International Telecommunication Union (ITU). (2023). *Measuring Digital Development: Facts and Figures 2023*. ITU Publications.
- Kizilcec, R. F., & Lee, H. (2022). Algorithmic fairness in education. In C. Rafferty & L. Page (Eds.), *The Ethics of Artificial Intelligence in Education* (pp. 174–202). Routledge.
- Kulik, J. A., & Fletcher, J. D. (2016). Effectiveness of intelligent tutoring systems: A meta-analytic review. *Review of Educational Research*, 86(1), 42–78.
- Levy, D. C. (2018). The decline of private higher education. *Higher Education Policy*, 26(1), 25–42.
- Mincer, J. (1974). *Schooling, Experience, and Earnings*. National Bureau of Economic Research.
- Rama, C. (2019). *La nueva educación universitaria digital en América Latina*. Universidade do Vale do Sapucaí.
- Rodríguez-Moreno, J., Agreda-Montoro, M., & Ortiz-Colón, A. M. (2022). Changes in teacher training within educational technology programs during the COVID-19 pandemic: A systematic review. *Education Sciences*, 12(2), 131.
- Sánchez-Prieto, J. C., Hernández-García, Á., García-Peñalvo, F. J., Chaparro-Peláez, J., & Olmos-Migueláñez, S. (2020). Break the walls! Second-order barriers and the acceptance of mLearning by first-year pre-service teachers. *Computers in Human Behavior*, 105, 106205.
- Scott, W. R. (2008). *Institutions and Organizations: Ideas and Interests* (3rd ed.). SAGE Publications.

Selwyn, N. (2022). The fate of (critical) educational technology studies in an AI age. *Learning, Media and Technology*, 47(2), 177–188.

Stevenson, M., & Phakiti, A. (2019). Automated feedback and second language writing. In K. Hyland & F. Hyland (Eds.), *Feedback in Second Language Writing: Contexts and Issues* (2nd ed., pp. 125–142). Cambridge University Press.

UNESCO-IESALC. (2021). *Higher Education and the 2030 Agenda: Advancing the Sustainable Development Goals*. UNESCO Institute for Higher Education in Latin America and the Caribbean.

VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychologist*, 46(4), 197–221.

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478.

Williamson, B. (2020). Making markets through digital platforms: Pearson, edu-business, and the (e)valuation of higher education. *Critical Studies in Education*, 62(1), 50–66.

Williamson, B., & Eynon, R. (2020). Historical threads, missing links, and future directions in AI in education. *Learning, Media and Technology*, 45(3), 223–235.

Wirtz, B. W., Weyerer, J. C., & Geyer, C. (2019). Artificial intelligence and the public sector: Applications and challenges. *International Journal of Public Administration*, 42(7), 596–615.

Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2019). Systematic review of research on artificial intelligence applications in higher education—where are the educators? *International Journal of Educational Technology in Higher Education*, 16(1), 39.